The Bounce-Back Effect:

What 130,000 Holes Tell Us About Momentum on the PGA Tour

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2020–2024 PGA Tour Data Analysis

ABSTRACT

Does making a birdie make you more likely to birdie the next hole? Most golf fans would say yes. The data disagrees. This study analyzes **130,499 consecutive hole pairs** from 246 players across 22 marquee PGA Tour tournaments (2020–2024). Using the regression framework of Arkes (2016), we find systematic **mean reversion** at the single-hole level: players who score below par on a hole tend to score slightly worse on the next, while players who score above par tend to score slightly better. Both effects are statistically significant. This pattern appears across nearly every player in the sample — 95% have negative streakiness scores. Streakiness does not predict tournament performance. The results suggest that at the elite level, emotional recalibration or strategic risk adjustment after extreme holes may be more powerful than momentum.

1. Does Momentum Exist in Golf?

Almost every golf broadcast has the same moment: a player birdies two holes in a row. The commentator says he's “in the zone.” The same is also true if a player makes two bogeys in a row. It just feels like his game is slipping from his grasp.

But is it? Or are we just very good at narrating patterns that don't exist?

This is the hot hand question, and it has generated decades of argument. In 1985, Gilovich, Vallone, and Tversky published what became one of the most cited papers in behavioral economics, analyzing NBA shooting records and concluding that the hot hand was a cognitive illusion. Fans saw streaks because humans are pattern-recognition machines, not because the streaks were real.

Then the consensus began to unravel. Better data and better methods started finding that maybe the hot hand was small, but real. A statistical bias in the original Gilovich study was identified in 2015 that suppressed the signal. Researchers found evidence for momentum in other sports. The debate is not settled.

Golf is one of the cleanest settings to test this. No defense, standardized scoring, 18 clear scoring units per round. And the psychological stakes are obvious and clear. Professional golf is five plus hours of sustained concentration constantly alternating between states of triumph and frustration.

> *"What does a birdie on the 7th hole tell us about the 8th? Statistically, probably less than we think."*

Arkes (2016) conducted the most rigorous golf test to date, using 11 years of ShotLink data and finding no hot hand, but a significant cold hand. He found that bad stretches predicted more bad holes, but good stretches couldn't sustain themselves. This study replicates his framework on 2020–2024 data and finds something different: not a cold hand, but a **bounce-back effect** (which is mean reversion in both directions) operating consistently across the field.

Dataset	**130,499**	*Consecutive hole-level observations, 2020–2024*
Players	**246**	*Unique PGA Tour players across 22 marquee tournaments*
Main finding	**Mean reversion**	*Both good and bad holes pull toward the average on the next hole*

2. The Data

The data comes from ESPN's public golf API, which is the same feed powering the ESPN app. No special access is required, just patience and rate-limiting. We pulled five seasons (2020–2024) covering three types of records:

- Tournament schedules for all PGA Tour events in each year
- Full leaderboards for every tournament (29,377 player-tournament rows)
- Hole-by-hole scoring for approximately 15 marquee events per year. This includes majors, flagship invitationals, and playoff events, collecting the top 30 finishers per event

Focusing on marquee events keeps data quality high. ESPN's coverage of The Masters or The Players is comprehensive; some smaller events have gaps. The top-30 filter gives a strong cross-section of competitive players per event without spending hours pulling data for every missed-cut field player.

After sorting by player → tournament → round → hole and creating lag variables, we filter to only truly consecutive holes (hole n immediately follows hole n-1 within the same round). This removes the first hole of every round from analysis, which is important — we don't want to accidentally measure whether a good finish on the back nine predicts a good start on the next day.

The final regression dataset is **130,499 consecutive hole pairs** across 246 players and 22 tournaments.

3. What a PGA Tour Round Actually Looks Like

Before testing for momentum, it helps to understand the baseline. What's the distribution of scores on a typical PGA Tour hole?

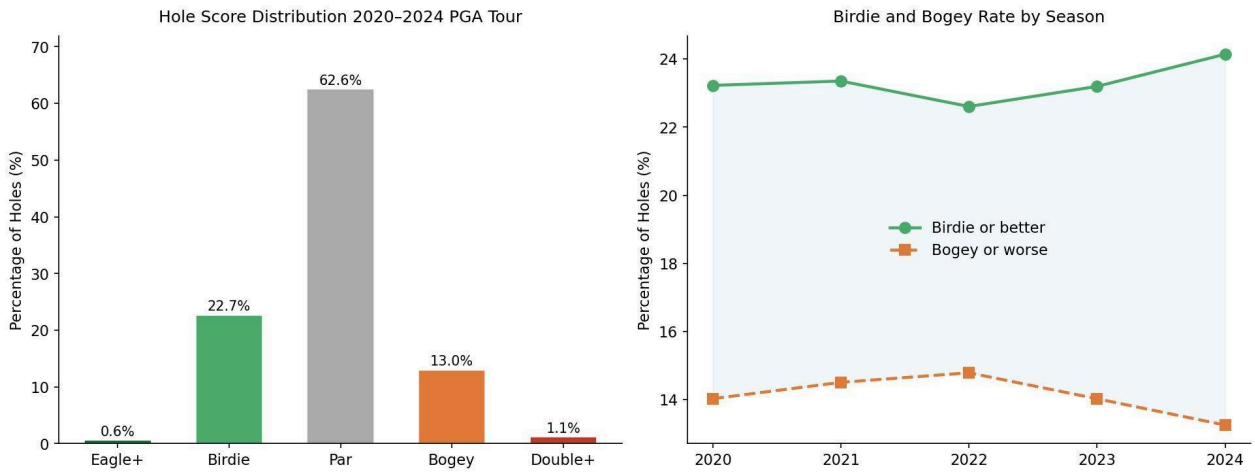

Figure 1. Score distribution (left) and birdie/bogey rate trends by year (right).

It comes as no surprise that the PGA Tour is elite. These are the best golfers in the world, and yet:

- 62.6% of holes are pars. The most common outcome, by far, is not making anything.
- 22.7% are birdies. Roughly 1 in 4 holes, these elite players score under par.
- 13.0% are bogeys. Even at the top of the game, a bad hole shows up once every 7-8 holes on average.
- Only 0.6% are eagles or better.

The trend line tells another story. Birdie rates have crept upward across the five seasons, from 23.2% in 2020 to 24.1% in 2024, while bogey rates dropped from 14.0% to 13.3%. The modern PGA Tour is getting better as training, equipment, and tactics improve.

4. What Comes After a Birdie?

In short, the research question was simply: if a player just made a birdie, are they more likely to birdie the next hole? The transition matrix shows the probability of each outcome given what happened on the previous hole.

If golf had no memory (each hole truly independent of the last) every row in the matrix would look the same.

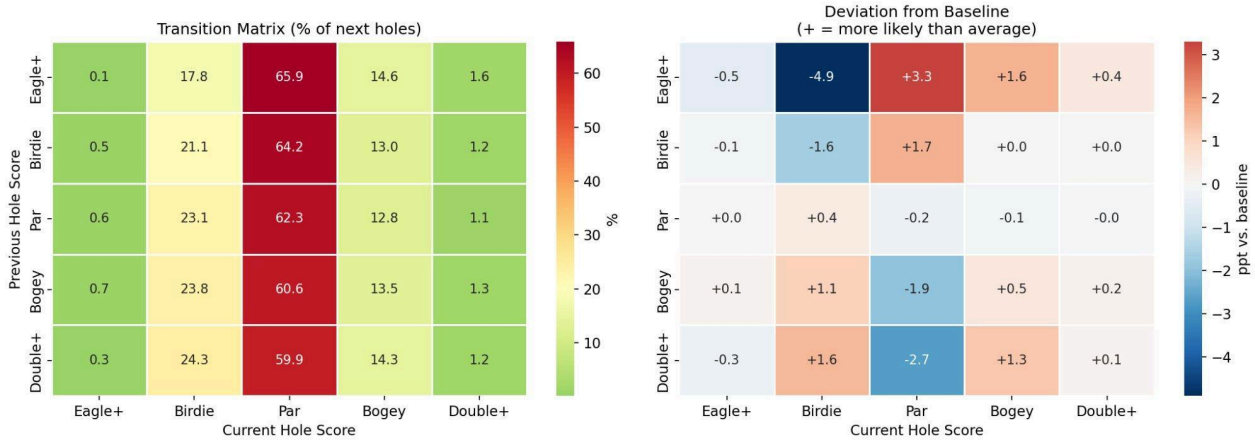

Figure 2. Transition matrix (left) and deviation from baseline rates (right). Blue = below average, Red = above average.

The deviations are small but consistent. The most striking number: after an Eagle or Birdie, players are

less likely to birdie the next hole than the overall average. After a Birdie, the next-hole birdie rate is –1.6 percentage points below baseline. After an Eagle, it's –0.5 ppt down.

The pattern reverses on the bogey side. After a Double bogey, players make birdies at a rate +1.6 ppt above baseline. After a Bogey, +1.1 ppt above baseline.

> *The data shows a gentle gravitational pull toward average. After a good hole, come back down. After a bad hole, bounce back up.*

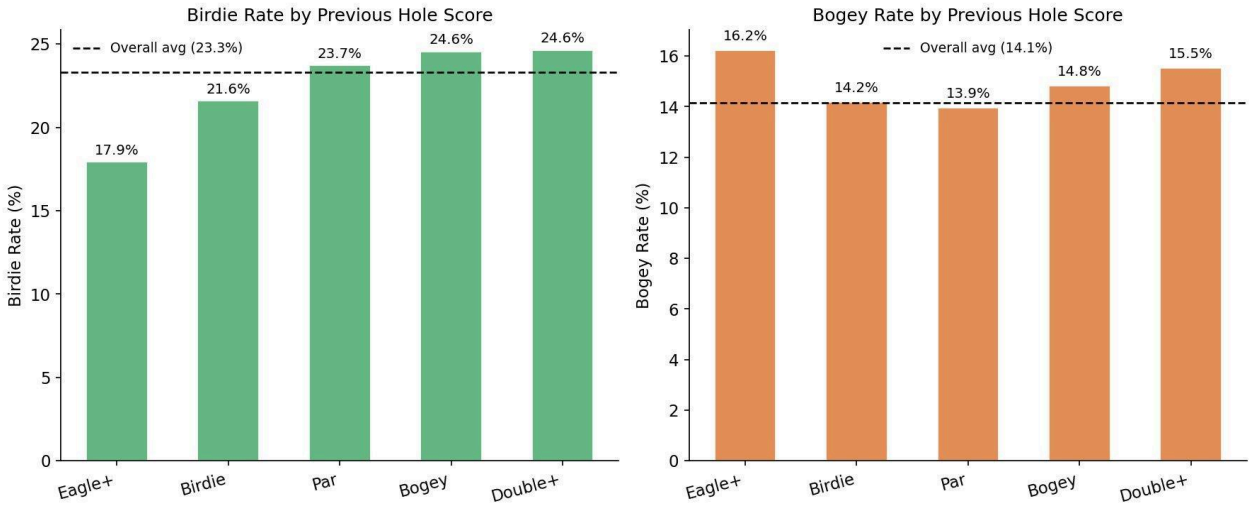

Figure 3. Birdie rates (left) and bogey rates (right) conditioned on previous hole score.

The conditional rates chart makes the pattern visible. After an Eagle, the birdie rate on the next hole drops to 17.9%, well below the 23.3% baseline. After a Double bogey, the bogey rate on the next hole (15.5%) is actually close to the overall average (14.1%), and the birdie rate is at its highest (24.6%).

This looks like mean reversion, not momentum. **But descriptive patterns can be misleading…**maybe harder holes simply follow easier ones on the course, or better players also happen to have had more recent birdies. We need to control for these.

5. Controlling for Hole Difficulty and Player Quality

The transition matrix is suggestive, but it doesn't prove anything on its own. Two obvious confounders could be driving the pattern:

- **Hole difficulty:** Maybe the 15th at Augusta just happens to follow the 14th, and both are hard. That's course design, not momentum. We control for this by computing the average score on each specific hole, in each round, at each tournament, across all players. When we say "the previous hole was a birdie," we mean it was a birdie *relative to how everyone else played that hole that day.*
- **Player quality:** Better players make more birdies. If Scottie Scheffler just birdied, the next hole might also be a birdie, not because of momentum, but because he's Scottie Scheffler. We control for this using each player's average score across the full dataset.

We run three versions of the same regression, each more controlled than the last:

- **Model 1 (Naive):** Just the raw relationship between last hole's score and this hole's score. No controls. This is the "TV commentator" model as it picks up everything, including all the confounders above.
- **Model 2 (Controlled):** Same as Model 1, but now we add the hole difficulty and player quality controls, which is a fairer test.
- **Model 3 (The Key Test):** This is where it gets interesting. Instead of treating "the previous hole was a birdie" and "the previous hole was a bogey" as two ends of the same spectrum, we split them into separate variables. One variable captures only how far *below* par the previous hole was. The other captures only how far *above* par it was. This matters because hot hand and cold hand are separate hypotheses. Maybe good runs lead to more good holes (hot hand) while bad runs just revert to average (no cold hand). Lumping them together into one variable would mask that.

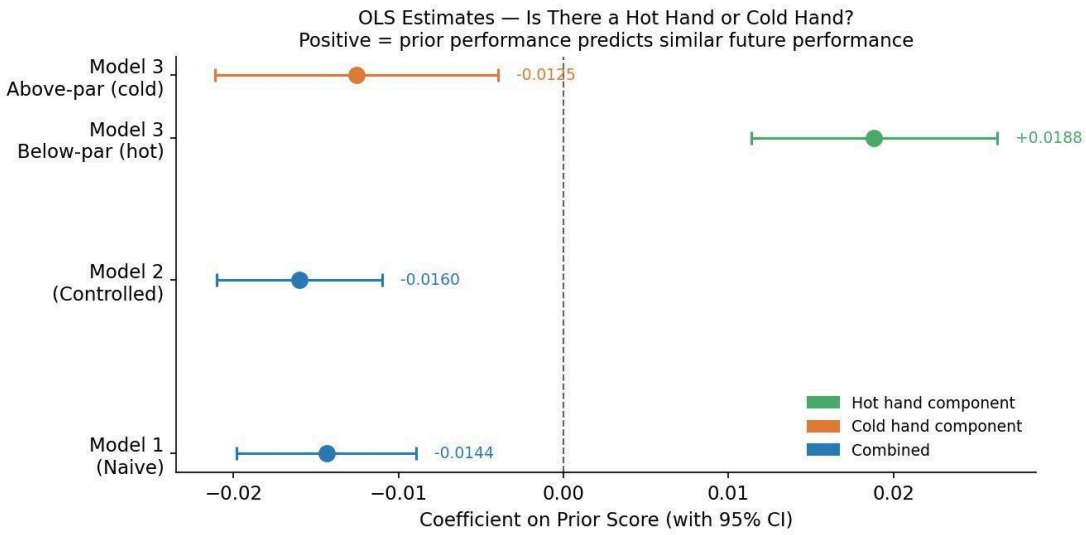

Figure 4. OLS coefficient estimates with 95% confidence intervals. All models use HC3 robust standard errors (N = 130,499). Figure 4 shows the coefficient estimates from all three models. Each dot is the estimated effect of the previous hole's score on the current hole, with the horizontal bars showing the 95% confidence interval. A dot to the right of zero means "good prior hole → better next hole" (momentum). A dot to the left means "good prior hole → worse next hole" (mean reversion).

In Models 1 and 2, the combined lag coefficient is around −0.014 to −0.016. The sign is already telling us the story: a better previous hole predicts a slightly worse next hole on average. But Model 2 barely moves from Model 1, which might seem odd given how much our controls matter (R^2 jumps from 0.0002 to 0.154). The reason is that the controls absorb *level* differences between players and holes. They explain why Scheffler scores better than average in general, but the autocorrelation of *deviations from that baseline* is a separate question.

Model 3 is the real test. The below-par component (green dot, right side of the plot) lands at +0.0188, positive, meaning a unit of below-par performance on the previous hole predicts a slightly *higher* (worse) score on the next hole. The above-par component (orange dot, left side) lands at −0.0125, negative, meaning a unit of above-par performance predicts a slightly *lower* (better) score next hole.

Both are statistically significant ($p < 0.001$ and $p = 0.004$ respectively)

Variable	Coefficient	p-value	Interpretation
Hot hand test (below-par component)	+0.0188	<0.001	Significant — mean reversion
Cold hand test (above-par component)	−0.0125	0.004	Significant — mean reversion
Arkes (2016) hot hand	+0.007	~0.3	Not significant
Arkes (2016) cold hand	+0.016–0.051	<0.001	Significant — cold hand

Both coefficients point toward mean reversion. A good prior hole pulls the next hole slightly up toward average. A bad prior hole pulls it slightly down toward average. This is not hot hand, cold hand, or any form of momentum: it is regression to the mean operating at the hole level.

The effects are small in absolute terms, about 0.013 to 0.019 strokes per unit. But they're consistent and highly significant across 130,000 observations.

How does this compare to Arkes (2016)? Arkes found no hot hand and a positive cold hand. Bad holes led to more bad holes. We find mean reversion in both directions: no cold hand, no hot hand, just a gravitational pull back to average. The most likely explanation is who we're looking at. Arkes used the top 200 players per year; we use top-30 finishers in marquee events. This is the most elite slice of an already elite field. It's plausible that at this level, post-bogey panic has been coached out of existence, and post-birdie overconfidence with it.

6. The Birdie-to-Birdie Question

The OLS regression treats score-to-par as a continuous number, which is fine statistically but slightly abstract. Let's ask the more intuitive question: after a birdie, how likely are you to birdie again?

To answer this we ran logistic regressions predicting the binary outcome, did the player make a birdie or not, based on whether the previous hole was a birdie or a bogey, controlling for hole difficulty and player quality.

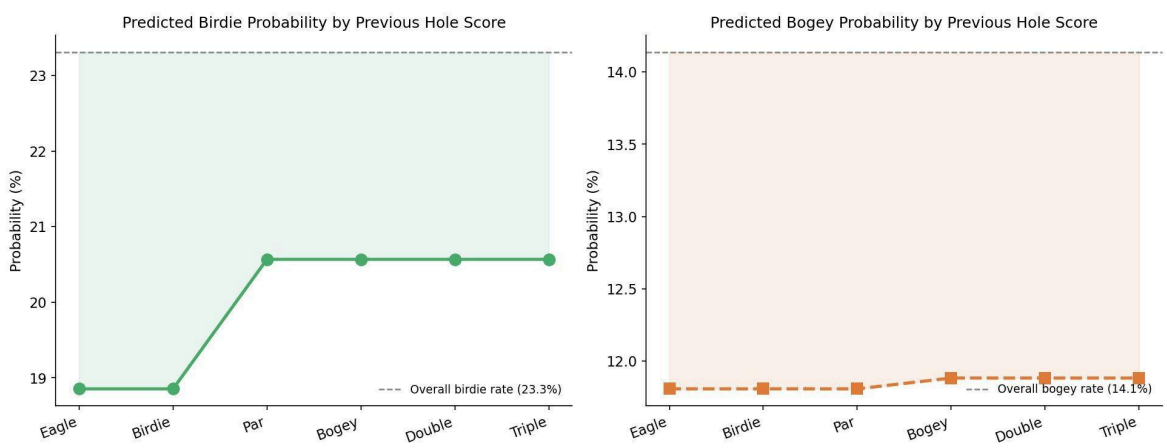

Figure 5. Predicted birdie probability (left) and bogey probability (right) by previous hole score, from logistic regression controlling for hole difficulty and player quality.

The logistic regression results are as follows:

Transition	Odds Ratio	p-value	What it means
After Birdie → Birdie	0.897	<0.001	~10% less likely to birdie
After Bogey → Bogey	1.007	0.760	No significant change
After Birdie → Bogey	1.029	0.139	No significant change
After Bogey → Birdie	1.136	<0.001	~14% more likely to birdie

The two significant results are directionally consistent with the OLS findings: after a birdie, you're slightly less likely to birdie again. After a bogey, you're more likely to birdie.

A few caveats are worth naming here. The binary "birdie or not / bogey or not" classification is a blunt instrument. An eagle and a tap-in birdie get treated identically, as do a lip-out par and a scrambled triple. The predicted probability curves in Figure 5 show this clearly: they're essentially flat from Par through Triple bogey because the model can't differentiate within the above-par or below-par buckets. A better version of this analysis would use a continuous prior score rather than binary flags, which is effectively what the OLS model does. The logistic results should be read as supporting evidence for the direction of the effect rather than precise estimates of its magnitude.

The broad finding holds: mean reversion, not momentum. A birdie does not beget a birdie.

7. Are Some Players Actually Streaky?

Everything above is an average across all players. But golf is an individual sport, and players have different temperaments. Maybe certain golfers genuinely ride momentum while others are bounce-back specialists.

We measure each player's "streakiness" as their average within-round lag-1 autocorrelation of hole scores. A positive score means their good holes cluster with good holes (streaky). A negative score means they tend to bounce back after extremes (mean-reverting).

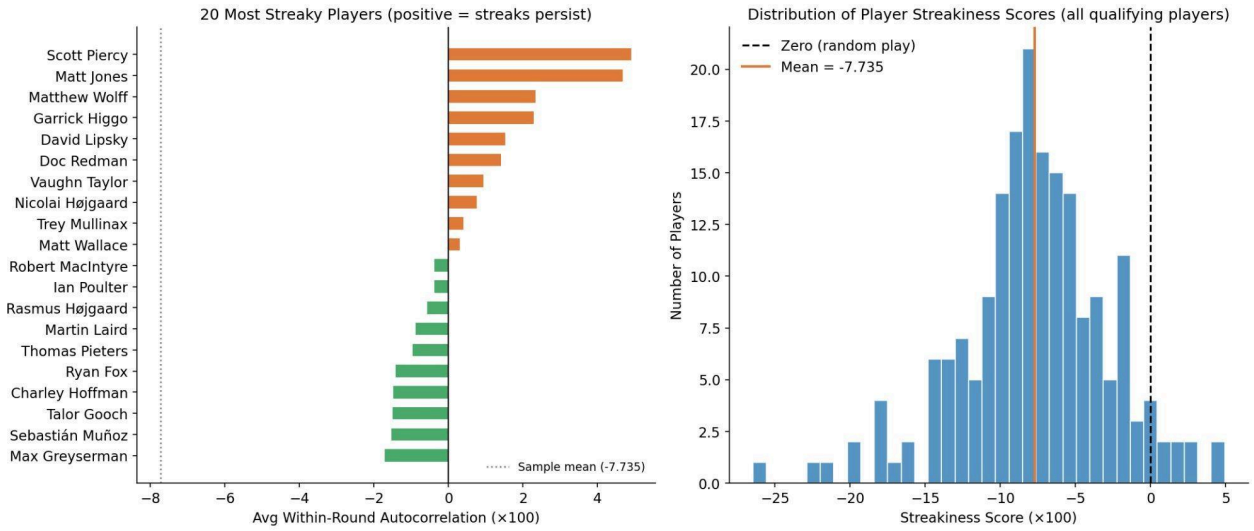

Figure 6. The 20 most streaky players by within-round autocorrelation (left), and the full distribution across 190 qualifying players (right).

The distribution is striking. Of 190 players with enough rounds to qualify:

- **180 players (95%)** have negative streakiness scores (mean-reverting)
- **Only 10 players (5%)** have positive scores (which we define as streaky)

The mean streakiness score is −0.077, and a one-sample t-test confirms this is highly significantly different from zero ($t = -21.49$, $p < 0.0001$). This is not a marginal effect. The PGA Tour field, as a whole, is strongly and consistently mean-reverting at the hole level.

The most streaky player in the sample is Scott Piercy (score: +0.049, 8 rounds), followed by Matt Jones (+0.047, 12 rounds) and Matthew Wolff (+0.024, 24 rounds). On the mean-reverting end: Romain Langasque (−0.265), Francesco Molinari (−0.222), and Xinjun Zhang (−0.219).

A few notable names from the scatter plot: Scottie Scheffler, Rory McIlroy, Jon Rahm, and Xander Schauffele all cluster in the −9 to −6 range; moderately to strongly mean-reverting. The best players in the world are also among the most likely to bounce back after a bad hole.

8. Does Being Streaky Help or Hurt?

If streakiness is a real playing style rather than statistical noise, it might predict tournament outcomes. A streaky player who can sustain a run of birdies might outscore a mean-reverting player of equal ability.

We merge each player's streakiness score with their average tournament score-to-par (minimum 5 tournament appearances) and run a simple regression.

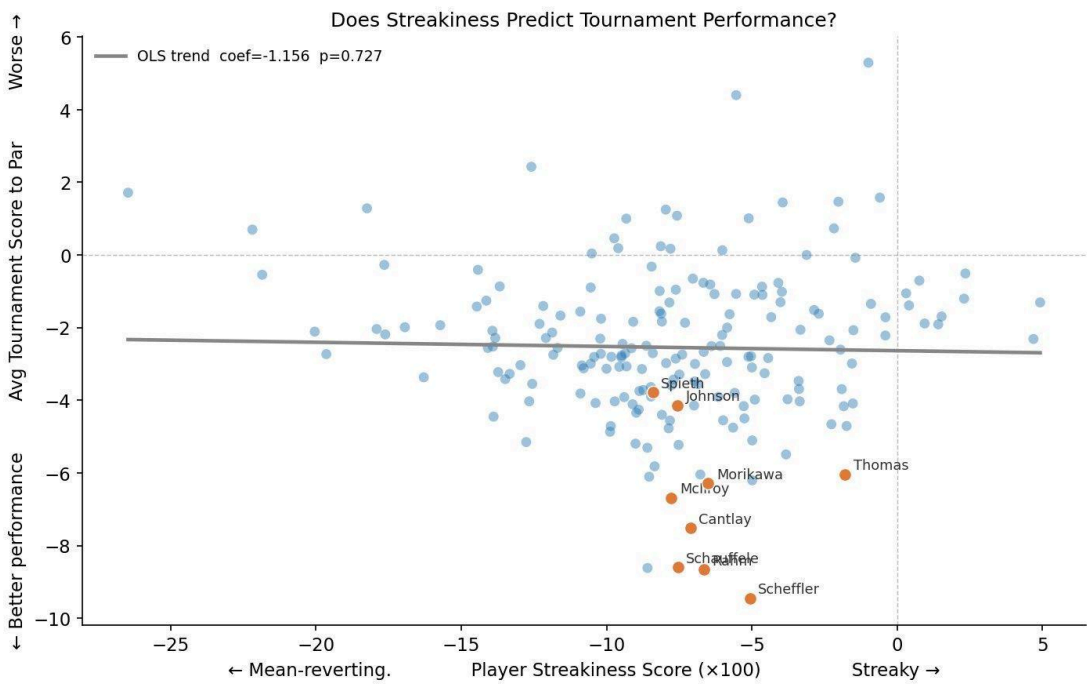

Figure 7. Player streakiness vs. average tournament score to par. Orange dots are labeled notable players. OLS trend line shown (not significant).

The regression coefficient is −1.156 ($p = 0.727$, $R^2 = 0.00071$). **Streakiness does not predict performance.** Whether a player is at the far mean-reverting end or the rare positive-streakiness end, their average score to par is similarly distributed.

This is important for interpretation. If mean reversion were simply a signature of elite-level skill — i.e., better players manage their games better and bounce back more — you'd expect a strong relationship. There isn't one. The players in the +2 strokes-per-tournament range are as scattered along the streakiness axis as the players at −8 or −9.

Streakiness appears to be an individual style that doesn't translate into scoring advantage or disadvantage. You can be a momentum player or a bounce-back player and perform equally well.

9. Why Would Golf Be Mean-Reverting?

The finding raises an obvious question: why? If birdie rates drop after a good hole, what's causing it? A few candidate explanations:

The Risk Adjustment Story

After making a birdie, a player might unconsciously (or consciously) become more conservative. They've built a cushion; now protect it. This means laying up when they might otherwise attack, putting to the center of the green rather than the pin. The slightly worse average outcome is a strategic choice, not a psychological breakdown.

This aligns with prospect theory (Kahneman & Tversky): being "up" relative to a reference point makes you risk-averse. A birdie moves you ahead of where you expected to be. You play not to lose it back.

After a bogey, the logic reverses. You're behind your reference point. The risk tolerance goes up. You attack pins you normally lay up to. Occasionally this leads to eagles and birdies; on average, the elevated aggression slightly improves your next score.

The Cognitive Reset Story

Golf requires unusual mental compartmentalization. Ben Hogan's maxim was that the game is a test of how few mistakes you make, not how brilliant your best shots are. Top players spend significant effort developing the mental routine to treat each hole as independent as to not carry the previous hole forward.

But "don't carry the last hole forward" may itself create mean reversion. After a birdie, the reset routine kicks in: don't get overconfident, stay in your process, play the next hole fresh. Paradoxically, the discipline to not get hot may produce the mean-reverting pattern we observe.

Why This Differs from Arkes (2016)

Arkes found a cold hand (bad holes lead to more bad holes) but no hot hand. We find mean reversion in both directions. Several things might explain the difference:

- **Sample composition:** Arkes used top-200 players per year from 2003–2013. We use top-30 finishers in marquee events from 2020–2024. Our sample skews even more elite and more toward the competitive end of performance.

- **Era effects:** The modern PGA Tour has embraced sports psychology and data analytics far more than 2003–2013. Tour players now have full mental performance coaching programs. The cold hand may have been coached away.

- **Sample size:** Arkes had ~2.8M holes; we have 130K. Our confidence intervals are wider, and our sample covers fewer players and events. The cold hand's absence in our data could partly reflect power limitations.

10. Conclusions

The narrative of momentum in golf is appealing. It makes broadcasts more exciting, it gives fans something to anticipate, and it matches how we intuitively experience sports. But five years of hole-by-hole PGA Tour data tells a different story.

The typical result on a PGA Tour hole is not what just happened. It's the long-run average. After a birdie, the next hole pulls back toward par. After a bogey, the next hole bounces back. This pattern is small in magnitude but statistically robust, appearing across nearly every player in the sample.

What this means practically:

- If you're watching a player go birdie-birdie, don't expect a third. The data slightly favors par next.
- If a player makes a double bogey, a bounce-back birdie is actually the most likely positive outcome. The stats support it.
- Whether a player is naturally "streaky" or naturally "mean-reverting" has no measurable effect on their average tournament performance. It's a style, not an advantage.

The deeper conclusion is about how top athletes manage extreme outcomes. In a game where the margin between a 64 and a 72 is a handful of shots across 18 holes, the ability to treat each hole as a fresh start, to not let a birdie become a hot hand or a bogey become a cold hand, may be one of the defining psychological features of elite performance.

> *The hot hand in golf doesn't disappear because players don't get hot. It disappears because the best players in the world are very, very good at coming back down.*

References

Arkes, J. (2010). Revisiting the hot hand theory with free throw data in a multivariate framework. Journal of Quantitative Analysis in Sports, 6(1).

Arkes, J. (2013). Misses in "hot hand" research. Journal of Sports Economics, 14(4), 401–410.

Arkes, J. (2016). The hot hand vs. cold hand on the PGA Tour. International Journal of Sport Finance, 11(2), 99–113.

Clark, R. D. (2003). Streakiness among professional golfers: Fact or fiction? International Journal of Sport Psychology, 34, 63–79.

Gilovich, T., Vallone, R., & Tversky, A. (1985). The hot hand in basketball: On the misperception of random sequences. Cognitive Psychology, 17, 295–314.

Kahneman, D. (2011). Thinking, Fast and Slow. Farrar, Straus and Giroux.

Livingston, J. A. (2012). The hot hand and the cold hand in professional golf. Journal of Economic Behavior & Organization, 81, 172–184.

Miller, J. B., & Sanjurjo, A. (2015). Surprised by the hot hand fallacy? A truth in the law of small numbers. Econometrica, 86(6), 2019–2047.

Stone, D., & Arkes, J. (2016). Reference points, prospect theory and momentum on the PGA Tour. Journal of Sports Economics, 17(7), 731–761.

Data sourced from ESPN's public golf API. Analysis code available in 01_data_collection.ipynb and 02_hot_hand_analysis.ipynb.